Simulation Assignment 6

A Multivariate Gaussian Target: RWM, MALA and GI-MALA

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1 Introduction

RWM and MALA are two basic implementations of the Metropolis–Hastings algorithm, with proposal distributions

$$q(y \mid x) = \mathcal{N}(y \mid x, 2\gamma\Sigma), \tag{1}$$
$$q(y \mid x) = \mathcal{N}(y \mid x + \gamma\Sigma\nabla \log \pi(x), 2\gamma\Sigma). \tag{2}$$

Here, x is the current position of the chain, y is the proposed new position, Σ is the preconditioning matrix, and γ is the step size. Equation (1) corresponds to RWM, while Equation (2) corresponds to MALA, which incorporates information from the gradient of the logarithm of the target density $\pi(x)$.

We define a new MCMC sampler based on the proposal distribution

$$q(y \mid x) = \mathcal{N}\big(y \mid x + \gamma\Sigma\nabla \log \pi(x), (2\gamma - \gamma^2)\Sigma\big). \tag{3}$$

The proposed sampler is called Gaussian Invariant MALA (GI-MALA), since for Gaussian targets $\pi(x) = \mathcal{N}(x \mid \mu, \Sigma)$, the proposal is reversible with respect to π, ensuring acceptance of all proposals, that is, $\alpha(x, y) = 1$. We assume that Σ is positive definite and $0 < \gamma < 2$, so that the proposal covariance is positive definite.

Proof of reversibility of GI-MALA for Gaussian targets

Since the target distribution is Gaussian,

$$\pi(x) = \mathcal{N}(x \mid \mu, \Sigma) \quad \Rightarrow \quad \nabla \log \pi(x) = -\Sigma^{-1}(x - \mu).$$

The GI-MALA proposal is therefore

$$q(y \mid x) = \mathcal{N}\big(y \mid (1 - \gamma)x + \gamma\mu, (2\gamma - \gamma^2)\Sigma\big).$$

Similarly,

$$q(x \mid y) = \mathcal{N}\big(x \mid (1 - \gamma)y + \gamma\mu, (2\gamma - \gamma^2)\Sigma\big).$$

To show that the proposal is reversible with respect to π, it is sufficient to show that

$$\pi(x)q(y \mid x) = \pi(y)q(x \mid y) \quad \Rightarrow \quad \alpha(x, y) = 1.$$

The corresponding probability densities are

$$\pi(x) = \frac{\exp\big[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\big]}{(2\pi)^{d/2}|\Sigma|^{1/2}},$$
$$q(y \mid x) = \frac{\exp\big[-\frac{1}{2b}(y - ax - \gamma\mu)^\top \Sigma^{-1}(y - ax - \gamma\mu)\big]}{(2\pi)^{d/2}b^{d/2}|\Sigma|^{1/2}},$$

where, for convenience, we set $a = 1 - \gamma$ and $b = 2\gamma - \gamma^2 = 1 - a^2$. The reverse densities have the same form with x and y interchanged.

In addition, we define $\delta_x = x - \mu$ and $\delta_y = y - \mu$. Taking logarithms of the ratio gives

$$\log \frac{\pi(x)q(y \mid x)}{\pi(y)q(x \mid y)} = -\frac{1}{2}\Big[\delta_x^\top \Sigma^{-1}\delta_x - \delta_y^\top \Sigma^{-1}\delta_y + \frac{1}{b}\big\{(\delta_y - a\delta_x)^\top \Sigma^{-1}(\delta_y - a\delta_x) - (\delta_x - a\delta_y)^\top \Sigma^{-1}(\delta_x - a\delta_y)\big\}\Big].$$

More explicitly,

$$(\delta_y - a\delta_x)^\top \Sigma^{-1}(\delta_y - a\delta_x) = \delta_y^\top \Sigma^{-1}\delta_y - 2a\delta_x^\top \Sigma^{-1}\delta_y + a^2\delta_x^\top \Sigma^{-1}\delta_x,$$
$$(\delta_x - a\delta_y)^\top \Sigma^{-1}(\delta_x - a\delta_y) = \delta_x^\top \Sigma^{-1}\delta_x - 2a\delta_y^\top \Sigma^{-1}\delta_x + a^2\delta_y^\top \Sigma^{-1}\delta_y.$$

Collecting the terms, the coefficients of the two quadratic forms are

$$1 + \frac{a^2 - 1}{b} = 0, \qquad -1 + \frac{1 - a^2}{b} = 0,$$

while $\delta_x^\top \Sigma^{-1}\delta_y = \delta_y^\top \Sigma^{-1}\delta_x$. Consequently,

$$\frac{\pi(x)q(y \mid x)}{\pi(y)q(x \mid y)} = \exp\left(-\frac{1}{2} \cdot 0\right) = 1.$$

Thus, we have verified reversibility of GI-MALA for Gaussian targets.

2 Assignment 6: A multivariate Gaussian target

In this example, we study the behaviour of the three MCMC algorithms in a known and fully controlled setting. We consider the d-dimensional Gaussian target

$$\pi(x) = \mathcal{N}(x \mid 0, \Sigma), \qquad \Sigma = \mathrm{diag}(s_1^2, \ldots, s_d^2), \qquad s_j = 0.01j,$$

with $d = 100$. Thus, the variance increases from 0.01^2 to 1^2 as the coordinate index increases.

1. **Comparison of the three algorithms.** Assume that Σ is known. Implement the three algorithms and experimentally find, for each algorithm, the value of γ that maximizes the minimum effective sample size (ESS) across all coordinates. Compare the ESS values and comment on the results.
2. **Construction of control variates.** For GI-MALA, find control variates for $F(x) = x$ and $F(x) = xx^\top$. Calculate the corresponding estimators and compare their variance with that of the ordinary sample average.

2.1 Question 1

2.1.1 RWM

The proposal distribution is

$$q(y \mid x) = \mathcal{N}(y \mid x, 2\gamma\Sigma),$$

with probability density

$$q(y \mid x) = \frac{1}{(2\pi)^{d/2}|2\gamma\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(y - x)^\top (2\gamma\Sigma)^{-1}(y - x)\right].$$

The reverse proposal has the same density, since the quadratic form is unchanged when x and y are interchanged. Therefore, the acceptance probability is

$$A = \min\left\{1, \frac{\pi(y)q(x \mid y)}{\pi(x)q(y \mid x)}\right\} = \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}.$$

Taking logarithms of the untruncated acceptance ratio r gives

$$\log r = -\frac{1}{2}\big(y^\top \Sigma^{-1} y - x^\top \Sigma^{-1} x\big), \qquad \log A = \min\{0, \log r\}.$$

2.1.2 MALA

The proposal distribution is

$$q(y \mid x) = \mathcal{N}(y \mid x + \gamma \Sigma \nabla \log \pi(x), 2\gamma\Sigma).$$

Since the target is Gaussian, $\nabla \log \pi(x) = -\Sigma^{-1}x$, so the proposal becomes

$$q(y \mid x) = \mathcal{N}(y \mid (1-\gamma)x, 2\gamma\Sigma).$$

Its density is

$$q(y \mid x) = \frac{\exp\left[-\frac{1}{4\gamma}(y-(1-\gamma)x)^\top \Sigma^{-1}(y-(1-\gamma)x)\right]}{(2\pi)^{d/2}|2\gamma\Sigma|^{1/2}},$$

and the reverse density is obtained by interchanging x and y. The acceptance probability is $A = \min\{1, r\}$, where

$$\begin{aligned}\log r = & -\frac{1}{2}\big(y^\top \Sigma^{-1} y - x^\top \Sigma^{-1} x\big) \\ & -\frac{1}{4\gamma}\Big[(x-(1-\gamma)y)^\top \Sigma^{-1}(x-(1-\gamma)y) - (y-(1-\gamma)x)^\top \Sigma^{-1}(y-(1-\gamma)x)\Big].\end{aligned}$$

Expanding the quadratic forms,

$$\begin{aligned}(y-(1-\gamma)x)^\top \Sigma^{-1}(y-(1-\gamma)x) &= y^\top \Sigma^{-1} y - 2(1-\gamma)x^\top \Sigma^{-1} y + (1-\gamma)^2 x^\top \Sigma^{-1} x, \\ (x-(1-\gamma)y)^\top \Sigma^{-1}(x-(1-\gamma)y) &= x^\top \Sigma^{-1} x - 2(1-\gamma)y^\top \Sigma^{-1} x + (1-\gamma)^2 y^\top \Sigma^{-1} y.\end{aligned}$$

Collecting the terms, the coefficient of $y^\top \Sigma^{-1} y$ is

$$-\frac{1}{2} + \frac{1-(1-\gamma)^2}{4\gamma} = -\frac{\gamma}{4},$$

and the coefficient of $x^\top \Sigma^{-1} x$ is

$$\frac{1}{2} - \frac{1-(1-\gamma)^2}{4\gamma} = \frac{\gamma}{4}.$$

The mixed terms cancel. Finally, we have

$$\log r = \frac{\gamma}{4}\big(x^\top \Sigma^{-1} x - y^\top \Sigma^{-1} y\big), \qquad \log A = \min\{0, \log r\}.$$

2.1.3 Gaussian Invariant MALA (GI-MALA)

The proposal distribution is

$$q(y \mid x) = \mathcal{N}\big(y \mid x + \gamma\Sigma\nabla \log \pi(x), (2\gamma - \gamma^2)\Sigma\big).$$

Since the target is Gaussian, this becomes

$$q(y \mid x) = \mathcal{N}\big(y \mid (1-\gamma)x, (2\gamma - \gamma^2)\Sigma\big), \qquad 0 < \gamma < 2.$$

Its probability density is

$$q(y \mid x) = \frac{\exp\left[-\frac{1}{2}(y-(1-\gamma)x)^\top \big((2\gamma-\gamma^2)\Sigma\big)^{-1}(y-(1-\gamma)x)\right]}{(2\pi)^{d/2}|(2\gamma-\gamma^2)\Sigma|^{1/2}}.$$

The reverse density is obtained by interchanging x and y. As shown in the introduction, the acceptance probability is equal to one.

Step size and ESS. For each coordinate, GI-MALA is an AR(1) process with coefficient $a = 1-\gamma$. At stationarity, the lag-k autocorrelation of X_j is a^k. Therefore, for estimating the coordinate mean and for large N at a fixed γ,

$$\frac{\mathrm{ESS}_j}{N} \approx \left(1 + 2\sum_{k=1}^{\infty} a^k\right)^{-1} = \frac{1-a}{1+a} = \frac{\gamma}{2-\gamma}.$$

For $\gamma = 1$, the proposal is the target itself and the generated draws are independent. This maximizes the population ESS for coordinate means within $0 < \gamma \leq 1$. However, over the full range $0 < \gamma < 2$, values larger than one introduce negative correlation and can give ESS larger than N. Thus, $\gamma = 1$ is not a global maximizer for this criterion. The expression increases as γ approaches two, although the endpoint itself is degenerate and is excluded. The large-N approximation is not uniform near that endpoint.

For the centred quadratic functions $X_jX_k - \Sigma_{jk}$, the lag-ℓ autocorrelation is $a^{2\ell}$. Hence,

$$\frac{\mathrm{ESS}_{jk}}{N} \approx \frac{1-a^2}{1+a^2},$$

which is maximized at $\gamma = 1$.

2.2 Question 2

2.2.1 $F(x) = x$

We require a control variate satisfying $\mathbb{E}_\pi[h(x)] = 0$ and highly correlated with F. We choose the gradient of the logarithm of the target density,

$$h(x) = \nabla \log \pi(x) = -\Sigma^{-1}x.$$

Since the target is $\mathcal{N}(0, \Sigma)$, we have $\mathbb{E}_\pi[h(x)] = 0$. Thus, the estimator with the control variate is

$$\widehat{\mu}_{\mathrm{CV}} = \frac{1}{N}\sum_{t=1}^{N}\big(X_t + Ch(X_t)\big),$$

where C is a matrix coefficient. For a single draw,

$$\begin{aligned}\operatorname{Cov}(X + Ch(X)) = {} & \operatorname{Cov}(X) + C \operatorname{Cov}(h(X))C^\top \\ & + \operatorname{Cov}(X, h(X))C^\top + C \operatorname{Cov}(h(X), X).\end{aligned}$$

For $X \sim \mathcal{N}(0, \Sigma)$,

$$\operatorname{Cov}(X, h(X)) = -I, \qquad \operatorname{Cov}(h(X)) = \Sigma^{-1}.$$

Taking $C = \Sigma$ eliminates the random term pointwise, since

$$X + \Sigma h(X) = X - \Sigma\Sigma^{-1}X = 0.$$

Finally,

$$\widehat{\mu}_{\mathrm{CV}} = 0, \qquad \operatorname{Cov}(\widehat{\mu}_{\mathrm{CV}}) = 0.$$

The cancellation holds for every sample, regardless of dependence between successive draws.

2.2.2 $F(x) = xx^\top$

The aim is to estimate $\mathbb{E}_\pi[xx^\top] = \Sigma$. Following the same reasoning, we first consider the Hessian of the logarithm of the target density,

$$H = \nabla^2 \log \pi(x) = -\Sigma^{-1}.$$

Since this is constant, we take instead

$$h(x) = \nabla \log \pi(x) \nabla \log \pi(x)^\top + \nabla^2 \log \pi(x).$$

For $\pi(x) = \mathcal{N}(x \mid 0, \Sigma)$, this becomes

$$h(x) = \Sigma^{-1}xx^\top\Sigma^{-1} - \Sigma^{-1}.$$

Thus, $\mathbb{E}_\pi[h(x)] = 0$, and the control is related to $xx^\top$. To eliminate the random term, we apply the transformation $-\Sigma h(x)\Sigma$. The estimator is

$$\begin{aligned}\widehat{\Sigma}_{\mathrm{CV}} &= \frac{1}{N}\sum_{t=1}^{N}\big(X_tX_t^\top - \Sigma h(X_t)\Sigma\big) \\ &= \frac{1}{N}\sum_{t=1}^{N}\big(X_tX_t^\top - (X_tX_t^\top - \Sigma)\big) = \Sigma.\end{aligned}$$

Consequently, every entry of the estimator has zero variance. Equivalently, we can choose $h_2(x) = xx^\top - \Sigma$ and subtract it directly.

2.2.3 Comparison with the ordinary sample average

For dependent MCMC draws, the variance of a sample average is not generally the marginal variance divided by N. For a scalar function f and a stationary chain,

$$\operatorname{Var}\left(\frac{1}{N}\sum_{t=1}^{N} f(X_t)\right) = \frac{1}{N^2}\left[N \operatorname{Var}_\pi(f(X)) + 2\sum_{\ell=1}^{N-1}(N-\ell)\operatorname{Cov}(f(X_1), f(X_{1+\ell}))\right].$$

For GI-MALA at $\gamma = 1$, the generated draws after the initial state are independent samples from the target. Using N such draws gives

$$\operatorname{Cov}(\overline{X}) = \frac{\Sigma}{N}, \qquad \operatorname{Var}\left(\frac{1}{N}\sum_{t=1}^{N} X_{t,j}X_{t,k}\right) = \frac{\Sigma_{jj}\Sigma_{kk} + \Sigma_{jk}^2}{N}.$$

In particular, since Σ is diagonal,

$$\mathrm{Var}(\overline{X}_j) = \frac{s_j^2}{N}, \qquad \mathrm{Var}(\overline{X_j^2}) = \frac{2s_j^4}{N}, \qquad \mathrm{Var}(\overline{X_j X_k}) = \frac{s_j^2 s_k^2}{N} \quad (j \neq k).$$

The corrected control-variate estimators have zero variance in all these cases. This exact cancellation is possible because the Gaussian target moments are already known; here it provides a controlled illustration of variance reduction.

3 Results

The three algorithms were applied to the 100-dimensional Gaussian target using the simplified experiment in `run_example.R`. One chain of 10,000 stored states was generated for each algorithm, including the initial state. Thus, each run performed 9,999 proposals. The initial value was 3 in every coordinate, and the random seed was 6210058. The step sizes were the same as those used in the original project:

- $\gamma = 3.5/d = 0.035$ for RWM;
- $\gamma = 30/d = 0.3$ for MALA;
- $\gamma = 1$ for GI-MALA.

For RWM, the first 5,000 stored states were removed. For MALA and GI-MALA, the first 1,000 were removed. All remaining states were used to calculate ESS with `coda::effectiveSize`; only the trace plots were subsampled, displaying every tenth stored state.

Metric	RWM	MALA	GI-MALA
Retained states	5,000	9,000	9,000
Minimum ESS	10.33	774.55	8,364.51
Average ESS	19.92	952.78	9,034.70
Acceptance rate	30.09%	56.10%	100%
Burn-in used	5,000	1,000	1,000
Step size γ	0.035	0.3	1

Table 1: Results of the simplified single-chain experiment. ESS is estimated separately for each of the 100 coordinates.

Table 1 records substantial differences in the performance of the three algorithms for this experiment.

- GI-MALA has ESS close to the 9,000 retained states. This is expected, since for $\gamma = 1$ each generated proposal is an independent draw from the target and is accepted. The estimated ESS can differ from the retained count; an estimated average slightly above 9,000 does not contradict independence.
- MALA has substantially larger ESS than RWM, illustrating the benefit of the gradient for mixing in this example.
- RWM has very low ESS despite retaining 5,000 states. The trace plot shows slow relaxation from the initial state, and this short run provides limited information about the target moments.

The initial value 3 is particularly far from the target for the first coordinate, whose standard deviation is 0.01. The shared vertical scale in the plots makes the behaviour of the smallest coordinates difficult to inspect. Therefore, the visual appearance alone does not establish convergence. No between-chain convergence diagnostic is reported, since this experiment uses one chain per algorithm. The acceptance rates include all proposals, including those made during burn-in; they are not estimates restricted to stationarity.

Step-size comparison. The assignment asks for the step size that maximizes

$$m(\gamma) = \min_{1 \le j \le d} \mathrm{ESS}_j(\gamma).$$

The table now includes this minimum, but the experiment evaluates only one chosen step size for each algorithm. Thus, it compares the selected values without establishing that they are optimal. A step-size search remains necessary to complete that part of Question 1. The retained counts also differ because different burn-in lengths were used; the ESS values describe these particular runs. As explained above, $\gamma = 1$ gives independent GI-MALA draws but does not maximize mean-estimation ESS over the whole range $0 < \gamma < 2$.

3.1 Control-variate results

For the 9,000 retained GI-MALA draws, the code calculates both the ordinary and controlled estimators. The largest absolute entry of the controlled mean was approximately 1.62×10^{-18}, and the largest absolute difference between the controlled second-moment estimator and Σ was approximately 2.22×10^{-16}. These are numerical rounding errors, consistent with the exact cancellations proved in Question 2.

Since the retained GI-MALA draws are independent, the ordinary estimator variances can be calculated exactly. For example, Thus, the controls remove the estimator variance completely in

Quantity estimated	Ordinary estimator variance	Controlled estimator variance
$\mathbb{E}[X_1]$	1.11×10^{-8}	0
$\mathbb{E}[X_{100}]$	1.11×10^{-4}	0
$\mathbb{E}[X_1^2]$	2.22×10^{-12}	0
$\mathbb{E}[X_{100}^2]$	2.22×10^{-4}	0
$\mathbb{E}[X_1 X_{100}]$	1.11×10^{-8}	0

Table 2: Theoretical estimator variances at $\gamma = 1$, using $N = 9{,}000$ independent draws. These are analytical variances, not empirical estimates from repeated runs.

this known Gaussian setting.

A Pseudocode

In the following algorithms, x denotes the current state. As in the R functions, I is the number of stored states, including the initial state. Store $x^{(0)}$ and then perform $I - 1$ transitions. The Metropolis algorithms distinguish the log ratio $\log r$ from the log acceptance probability $\log A$.

Algorithm 1: RWM for a d-dimensional Gaussian target

Inputs: initial state $x^{(0)}$, number of stored states $I \ge 2$, step size $\gamma > 0$, covariance matrix Σ.

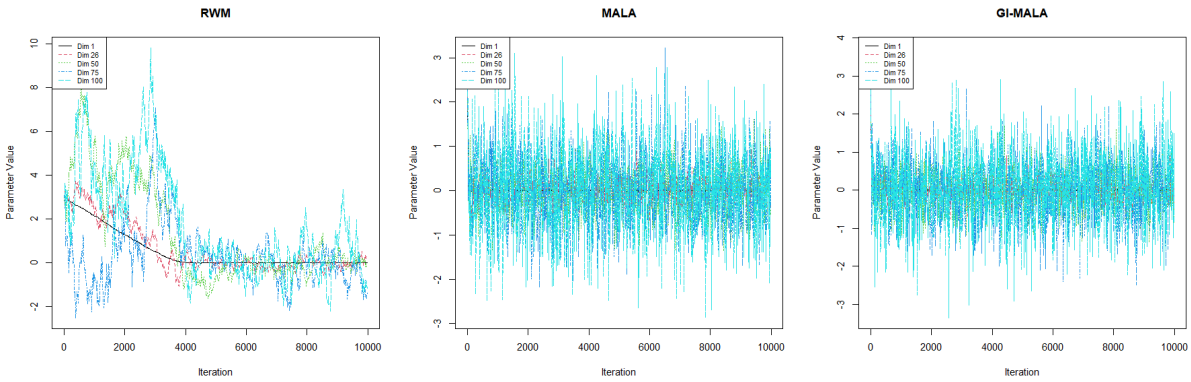


Figure 1: Trace plots from the simplified experiment: RWM (left), MALA (middle) and GI-MALA (right). Each chain starts at 3 in every coordinate. Only every tenth stored state is plotted; the ESS calculation uses all retained states.

1. Set $x = x^{(0)}$.
2. For $t = 1, \ldots, I-1$:
   (a) Propose $y \sim \mathcal{N}(x, 2\gamma\Sigma)$.
   (b) Compute $\log r = -\frac{1}{2}(y^\top \Sigma^{-1} y - x^\top \Sigma^{-1} x)$ and $\log A = \min(0, \log r)$.
   (c) Draw $U \sim \text{Uniform}(0, 1)$.
   (d) If $\log U < \log A$, set $x = y$; otherwise retain x.
   (e) Store $x^{(t)} = x$.

Algorithm 2: MALA for a d-dimensional Gaussian target

Inputs: initial state $x^{(0)}$, number of stored states $I \geq 2$, step size $\gamma > 0$, covariance matrix Σ.

1. Set $x = x^{(0)}$.
2. For $t = 1, \ldots, I-1$:
   (a) Propose $y \sim \mathcal{N}((1-\gamma)x, 2\gamma\Sigma)$.
   (b) Compute $\log r = \frac{\gamma}{4}(x^\top \Sigma^{-1} x - y^\top \Sigma^{-1} y)$ and $\log A = \min(0, \log r)$.
   (c) Draw $U \sim \text{Uniform}(0, 1)$.
   (d) If $\log U < \log A$, set $x = y$; otherwise retain x.
   (e) Store $x^{(t)} = x$.

Algorithm 3: GI-MALA for a d-dimensional Gaussian target

Inputs: initial state $x^{(0)}$, number of stored states $I \geq 2$, step size $0 < \gamma < 2$, covariance matrix Σ.

1. For $t = 1, \ldots, I-1$:
   (a) Propose $y \sim \mathcal{N}((1-\gamma)x^{(t-1)}, (2\gamma - \gamma^2)\Sigma)$.
   (b) The acceptance probability is $A = 1$.
   (c) Accept the proposal and store $x^{(t)} = y$.